



Implementing cross-border transfer of domestic plasma to obtain plasma-derived medicinal products policy guidance



**World Health
Organization**

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Preface

Based on various concerns, governmental authorities in low- and middle-income countries, as well as in some high-income countries, often ban exportation of domestic human plasma, or do not have in place regulations and legislation to guide the export of plasma. However, the World Health Organization (WHO) regards transfer of domestic plasma across borders, when linked to provision in return of industrially manufactured plasma-derived medicinal products (PDMPs), as a compelling national interest. This is especially so in settings where access to essential PDMPs is inadequate to meet patient needs. This document was developed to inform ministries of health about the WHO policy as part of a programme to improve global access to essential PDMPs. It discusses the existing barriers to exportation of plasma and provides the rationale for policy decisions in favour of foreign processing (“fractionation”) of domestic plasma to improve availability and affordability of PDMPs. It also recommends measures for overcoming barriers to cross-border transfer of domestic plasma, which is defined as shipment of domestic plasma to a manufacturer in a foreign country under an arrangement for “contract” or “toll” fractionation.



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Abbreviations and acronyms

AABB	Association for the Advancement of Blood and Biotherapies
AFSBT	African Society of Blood Transfusion
EU/EEA	European Union/European Economic Area
GDP	Good Delivery Practice
GMP	Good Manufacturing Practice
HCV	hepatitis C virus
IPFA	International Plasma and Fractionation Association
IQPP	International Quality Plasma Programme
ISBT	International Society of Blood Transfusion
LMICs	low- and middle-income countries
MA	marketing authorization
MAH	marketing authorization holder
MCA	medicines control agency
NAT	nucleic acid testing
NCL	national control laboratory
NRA	national regulatory authority
OMCL	official medicines control laboratory
OCABR	official control authority batch release
PDMPs	plasma-derived medicinal products
PPTA	Plasma Protein Therapeutics Association
SoHO	substances of human origin
WHO	World Health Organization
US FDA	United States Food and Drug Administration
ZaZiBoNa	Collaborative medicines registration initiative of countries in Southern Africa that is focused on dossier assessments and current Good Manufacturing Practices (cGMP) inspections

Glossary

Achilles project	WHO project initiated in 2009 aimed at assisting Member States to provide an affordable and sustainable supply of essential, high-quality blood-derived medicines to their populations.
“Contract” or “toll” plasma fractionation	Arrangement in which a foreign pharmaceutical company is contracted to process (i.e. fractionate) domestically collected plasma into medicinal products intended to be returned for use within the country that collected the domestic plasma. Toll fractionation is sometimes understood to be the type of contract fractionation in which the returned PDMPs are made solely from the transferred domestic plasma.
Cross-border transfer of domestic plasma	Shipment of domestic plasma to a manufacturer in a foreign country under an arrangement for “contract” or “toll” fractionation.
Financial neutrality	The principle, under the “Oviedo” Convention on Human Rights and Biomedicine, that “the human body and its parts shall not give rise to financial gain”. In this document the principle is interpreted to consider a) reimbursement of expenses incurred and b) compensation for effort made (for example, time and inconvenience) as deemed acceptable.
Plasma fractionation	The industrial process of separating plasma into various therapeutic protein components, typically known as plasma-derived medicinal products.
Recovered plasma	Plasma obtained by centrifugation of whole blood donations that is dedicated for manufacturing of medicinal products.
Source plasma	Plasma obtained by plasmapheresis that is dedicated for manufacturing of medicinal products.



Executive summary

Lack of availability of essential plasma-derived medicinal products (PDMPs) in low- and middle-income countries (LMICs) results in otherwise preventable morbidity and mortality in patients with hereditary and acquired deficiencies in plasma proteins, which occur in all populations. The impact of this inadequacy is especially great for patients with coagulation disorders and those with immune deficiencies who require replacement with clotting factor concentrates and immunoglobulins, respectively. Insufficient access to industrially prepared (fractionated) plasma products and, when applicable, their alternative recombinant products, is generally attributed to a global shortage of plasma suitable for fractionation combined with high costs of manufacturing. Under the “Achilles project” the World Health Organization (WHO) seeks to assist Member States to provide their populations with an affordable and sustainable supply of essential PDMPs. This is achieved through strengthening their blood systems to enable foreign or, where feasible, domestic fractionation of quality-assured domestic plasma. In the short term, and typically also in the long term, contracting for foreign fractionation of domestic plasma is the most realistic approach to providing sustainable access to a basic supply of essential PDMPs in LMICs. This is because of the high costs and technological challenges of developing a domestic fractionation plant and the requirement, often unattainable, of an annual volume of suitably quality-assured domestic plasma large enough to permit economically efficient and attractive manufacturing.

Fundamentally, foreign fractionation of domestic plasma entails shipment of plasma to a manufacturing facility in a foreign country, which can raise concerns about plasma exportation or trans-border transfer of plasma. However, in this document, a distinction is made between “exportation” per se of plasma, which typically means only sale to a foreign entity (which while adding to the global supply of plasma protein therapies may have no direct link to a benefit for local patients in the plasma-producing country) and “cross-border transfer of domestic plasma” defined as shipments taking place within a well-controlled national plasma programme that assures return of PDMPs manufactured solely or in part from that plasma. The document provides recommendations on legal, regulatory and policy conditions for assuring the quality of plasma for fractionation and protecting the health of plasma donors. It also discusses alternative arrangements for foreign fractionation of domestic plasma (termed “contract” or “toll” fractionation), which realize the medical and economic value of the plasma as a national resource for public health. These recommendations, which reflect common practices in high-income countries, address and mitigate the ethical, scientific, economic and practical concerns that underly existing legal and policy barriers to cross-border domestic plasma transfer and balance them against the aim of improving patient access to essential PDMPs. Based on its analysis of the issues, WHO recommends the following conditions for national acceptance of cross-border transfer of domestic plasma:

- A competent national governmental authority provides oversight to assure that Good Manufacturing Practices are followed at all levels of the blood and plasma system.
- Regulatory safeguards are in place to assure the health and safety of donors, including limitations on the volume and frequency of collection of blood and plasma.
- Legal requirements backed by adequate enforcement mechanisms are in place to prevent importation of compromised or counterfeit PDMPs.
- If PDMPs imported under the programme involve use of non-domestic plasma solely or in a mixture with domestic plasma, a scientific and regulatory assessment has been made of the acceptability of the final products, based on comparable risks of the differently sourced plasmas.

- Where a programme of plasmapheresis exists to generate source plasma, measures are in place to avoid compensation in excess of financial neutrality.
- The value of the plasma that is exported as raw material partly offsets the costs of the finished PDMPs.
- Donors are informed that PDMPs manufactured from their donations that have costs of preparation and procurement will become economically valued products for some patients' health and survival.
- A national programme exists to assure equitable patient access to the PDMPs obtained through contract or toll fractionation.
- Cost reimbursement for procurement of PDMPs under contract or toll fractionation is managed under a sustainable national or regional budgetary commitment.



CHAPTER 1

Introduction

One of the main challenges identified as hampering global efforts to address the current barriers to access to safe blood and blood products is the lack of availability of plasma-derived medicinal products (PDMPs). The World Health Organization (WHO) Action Framework to advance universal access to safe, effective and quality-assured blood products 2020–2023, proposed six strategic objectives (1). One of the outcomes of strategic objective three “Functioning and efficiently managed blood services” is to increase the volume and quality of plasma available for PDMPs.

The purpose of this Policy guidance is to provide a global strategy for utilizing recovered plasma that is not used for transfusion, as well as source plasma, for the production of PDMPs through “contract” (or “toll”) plasma fractionation. This document first identifies the concerns that underlie prohibitions against the cross-border transfer of domestic plasma. This is followed by definitions of the conditions under which this practice could be deemed appropriate in low- and middle-income countries (LMICs) in a manner consistent with good stewardship of national resources and attention to individual and public health. In line with the resolutions of the World Health Assembly, WHO encourages the collection of plasma from voluntary non-remunerated donors (2). Plasma donation should preserve “financial neutrality”. This is consistent with the principles for promoting ethical practices in the donation and management of medical products of human origin. These were presented in the report prepared by the WHO Secretariat for the Seventieth World Health Assembly at the request of WHO’s Member States, as contained in document A70/19, which was noted by the World Health Assembly, WHO’s supreme governing body (3). The rationale for developing is as follows: WHO recognizes the value of a stepwise approach in LMICs. This approach enables eventual access to PDMPs through local preparation of safe plasma proteins, including use of a validated method of virus inactivation. As local preparation of these products in accordance with Good Manufacturing Practices (GMP) expands, the volume of quality-assured plasma potentially suitable for fractionation increases. When a country consistently produces >30 000–50 000 litres of quality-assured plasma per year, which also meets international standards for fractionation including competent regulatory oversight, the opportunity arises to consider engaging in an economic arrangement with a foreign fractionator. Under such an arrangement, domestic plasma is exported for fractionation, added to the manufacturer’s plasma pool, and specified PDMPs made in part from the transferred plasma are subsequently imported to the countries that supplied the plasma. See section 7.2 for more information on alternative arrangements for foreign fractionation of domestic plasma. When large volumes are collected, a type of contract fractionation can be arranged where the imported PDMPs are made solely from the plasma that was exported.

Although there are many variations on the arrangement for plasma fractionation under such a foreign contract, a common barrier is resistance to cross-border transfer of domestic plasma. This resistance is grounded in ethical, economic and scientific concerns that can be addressed in a well-controlled national plasma programme.

The following chapters discuss the background and the reasons for resistance and propose solutions.

Development of the document

This document was developed based on identified issues predominantly in low- and middle-income countries regarding underlying concerns on prohibitions against the cross-border transfer of domestic plasma and the existing barriers to the exportation of domestic plasma to improve the availability and affordability of PDMPs. An initial virtual meeting of a selected group of experts in transfusion medicine and plasma fractionation, WHO Regional Advisers for Blood and Transplantation, and WHO headquarter Blood and Other Products of Human Origin staff, was initiated to discuss these issues resulting in agreement on the need to develop policy guidance on the implementation of cross-border transfer of domestic plasma to obtain plasma-derived medicinal products (PDMPs) was discussed and established. The group identified the key topic areas and individual working groups were established to produce the initial text for each of these areas. An editorial group was formed from the leaders of the individual working groups and the final document developed through a reiterative process by the initial group before review by a number of WHO identified external experts.



CHAPTER 2

Background

2.1 PDMPs are needed everywhere, regardless of income

Industrially manufactured (“fractionated”) PDMPs, including coagulation factor concentrates FVIII and FIX and normal immunoglobulin, are listed in the WHO Model List of Essential Medicines for adults and for children. These lists identify the medicines essential to meet a population’s health care needs. A number of LMICs have little or no access to PDMPs, primarily because of their very high commercial costs relative to the available resources, and the lack of plasma collected in these regions or the inability to meet GMP requirements. The inability to provide these medicines has a profoundly negative impact on the morbidity and mortality of people born with inherited disorders related to lack of, or non-functional forms of plasma proteins. These include primary immunodeficiency, haemophilia A, haemophilia B, von Willebrand Disease, alpha-1 antitrypsin deficiency, hereditary angioedema and other life-threatening rare diseases of unknown (autoimmune or inflammatory) cause, such as idiopathic thrombocytopenic purpura, Kawasaki disease and chronic inflammatory demyelinating polyneuropathy. People with these conditions are found in all populations, whether or not they are identified diagnostically within the health system.

2.2 Manufacturing process of PDMPs

PDMPs are manufactured in a process that takes up to 12 months. Internationally recognized standards for preparation of plasma for fractionation, such as those from WHO, should be followed (4). The process starts with the donation of plasma. This may be plasma collected through apheresis or recovered plasma, which is plasma not needed for transfusion, that is separated from a whole blood donation after the donation process is completed.

Following collection, each donated unit of plasma is tested for suitability for further manufacture and frozen. Depending on the fractionator, donated plasma may be held for as long as 60 days prior to pooling to permit retrieval if the donor tests positive for a transfusion-transmitted infection during the hold period. Plasma may be subjected to additional testing (for example, nucleic acid test (NAT)) confirming non-reactivity for relevant viruses. Each plasma donation must pass these rigorous steps to be pooled with other plasma donations. Each plasma pool is further tested with NAT for specific transfusion-transmitted virus infections before fractionation begins. During this process, life-saving proteins

are extracted. The separated, specific proteins are purified, virally-reduced and adjusted to the optimal composition, concentration and dosage before becoming the finished therapeutic product. Each plasma donation, whether source or recovered, is tracked throughout the process – from donation all the way to the final products. Each year approximately 130 source plasma donations are needed to treat one person with a primary immunodeficiency, 900 donations to treat one person with alpha-1 antitrypsin deficiency and 1200 donations to treat one person with haemophilia (5). For economic reasons, a minimum annual volume of approximately >30 000 litres of quality-assured plasma is needed to make contract fractionation worthwhile. However, the volume is subject to agreement with the fractionator.

2.3 Previous and ongoing efforts to ensure access to PDMPs in LMICs

The supply of PDMPs is lacking or restricted in many countries, and, over the years, many attempts have been made to improve the situation. Some countries have made plans to rectify the lack of products by setting up domestic fractionation facilities. However, very few countries have succeeded with such projects for various reasons. These include lack of funding and resources, and lack of knowledge or basic infrastructure to perform the complete plasma-to-final-product-process, such as that related to meeting high quality demands.

In 2009, under the “Achilles project” WHO recognized that insufficient production of plasma meeting standards for industrial fractionation was a limiting factor in global access to PDMPs. WHO revitalized the Achilles project under the “Action Framework to advance universal access to safe, effective and quality-assured blood products 2020–2023” and published its *Guidance on increasing supplies of plasma-derived medicinal products in low- and middle-income countries through fractionation of domestic plasma* in 2021 (1,6). In this guidance, WHO recognized the value of a stepwise approach in LMICs towards eventual access to PDMPs. These steps include:

- the vitally important step of setting up a quality system of a sufficient level in the blood system, for collection of plasma;
- the possibility to start with local preparation, in quality-compliant blood centres, of safe plasma proteins including virally-inactivated cryoprecipitates; and
- most importantly for this guidance, the possibility to utilize domestically collected plasma for fractionation by an international partner to achieve access to the PDMPs.

All these steps need to be considered before any plans for investments in local fractionation are made. When local capacity to implement GMP expands and when GMP is actually implemented, the volume of quality-assured plasma potentially suitable for fractionation increases.

2.4 Existing barriers to cross-border transfer of plasma

When a country considers engaging in a contractual and economic arrangement with a foreign fractionator, the domestically collected plasma has to be transferred to the country or countries where the manufacturing facilities are located. There are many possible arrangements for plasma fractionation under a foreign contract under which domestic plasma is transferred for fractionation. The specified PDMPs needed in the plasma-supplying country can be produced by a fractionator from the transferred domestic plasma, in combination with non-domestic plasma, or solely

from a corresponding amount of non-domestic plasma. Subsequently, the PDMPs are imported with the appropriate authorization for clinical use in that country.

In blood systems where transfer of plasma is a new step in the country's handling of blood and plasma, there can be initial resistance to the idea, often based on perceived ethical, economic or scientific concerns, or functional impediments. Such resistance creates barriers to the use of plasma for further manufacturing outside a country's borders (7).

Barriers to plasma transfer can be placed into two general categories: concerns of the country considering plasma transfer, and strict requirements of the fractionator. However, such concerns can be addressed in a well-controlled national plasma programme that includes a proper regulatory and legislative framework and appropriate local (national or regional) policies, or through public–private partnerships.

Concerns of the country considering plasma transfer

Typical concerns of the country considering plasma transfer are:

- lack of national regulation, policies or a legislative framework, including competent oversight by the responsible authority, which would allow cross-border transfer or export of domestic plasma for this purpose;
- economic considerations regarding allocation of resources across the health care system;
- ethics of donation for plasma transfer;
- concerns about donor health arising from the need to collect more plasma; and
- lack of information or analysis of which model is most appropriate in the national context.

These elements will be further explored in this document, which endeavours to provide evidence to alleviate those concerns and provide a roadmap for building the required frameworks.

Barriers to acceptability by the fractionator

These barriers fall into two categories – inadequate volume and insufficient quality – as described in WHO's 2021 *Guidance on increasing supplies of plasma-derived medicinal products in low- and middle-income countries through fractionation of domestic plasma* (6).

The volume of plasma required is not defined and varies by fractionator, but it can be difficult to incorporate smaller volumes of plasma as independent fractionation pools into the manufacturing schedule or to do so in a way that is economically advantageous to the collecting country. Increasing the volume of plasma collected is generally required of a blood system embarking on the process of plasma transfer.

Factors preventing the collection of adequate volumes include:

- insufficient equipment capable of freezing within the specified time frame in addition to maintaining the cold chain including plasma at temperatures of -20°C and colder, depending on legislation;

- inadequate transport infrastructure in the country, which would hinder or delay the movement of plasma to a foreign fractionator;
- fragmentation within the blood system, preventing the considerable costs of maintaining and certifying GMP compliance from being shared among donation centres;
- lack of harmonized national regulation, preventing plasma collected at individual centres from being pooled; and
- focus on the significant upfront cost of initiating a plasmapheresis programme, which has been found to be a key feature of a system collecting an adequate volume of plasma, rather than the efficacy and sustainability of supply of PDMPs through this means.

Quality plasma is a requirement for fractionators, all of whom must comply with GMP throughout collection and processing to ensure safety, quality and manufacturing consistency to maintain their marketing authorization (MA). Failure to comply with specific requirements can lead to inadequate quality and safety and raise questions about the suitability of therapies with these final products for patients' use.

These specific requirements include practices and documentation of:

- suitable design and layout for safe and efficient function;
- hiring and maintaining trained and qualified personnel;
- recruiting and maintaining qualified and regular donors;
- checks on donor eligibility;
- implementation of safety measures for blood and plasma collection;
- testing donations for transfusion-transmissible infectious diseases;
- use of validated equipment, reagents, medical devices and procedures;
- timely collection and standardized freezing conditions;
- appropriate plasma storage and transport facilities and conditions;
- periodic quality control testing; and
- bi-directional traceability between donations and end-products (4, 8).

Furthermore, fractionators require information on the risks posed by transfusion-transmitted viruses in the donor population and the mitigation measures in place. Specifically, this should include NAT-based screening for HIV, hepatitis B and hepatitis C as a minimum. Fractionators need to consider the residual risks after mitigation measures to ensure that the processes employed by the system to inactivate and/or remove these viruses will be sufficient to ensure infectious safety of the end-products.

Fractionators assess many elements when considering entering into a contract. They may vary between contractors and go beyond those listed here. It is critical that blood systems work closely with the contract fractionator from the beginning, to identify gaps and ensure that the provided plasma will be acceptable.

2.5 National experiences

Based on WHO global survey data, 60 out of 108 countries, 34 of which are in the WHO African Region, reported annual whole blood collection rates less than 10 donations per 1000 people for the year 2018.¹ Only a few African countries are using the surplus plasma recovered from whole blood donations to produce much-needed PDMPs. Most African countries have an established national blood policy in place. According to this policy the national blood establishments, together with regional, satellite and mobile collection centres, are responsible for collecting, testing, component separation, and supplying blood and blood components to both public and private health care facilities to meet patient needs. Most of the blood establishments are wholly owned, financed and managed by the national government. Over the past two decades, blood collections in Africa have increased significantly, in part due to international funding from organizations such as the US President's Emergency Plan for AIDS Relief, advertising and donor recruitment activities, as well as guidelines and assistance from organizations such as WHO (9). To improve the quality and safety of blood and blood products and to ensure compliance with international standards, many countries have sought accreditation from independent organizations such as the African Society of Blood Transfusion (AfSBT) and the Association for the Advancement of Blood and Biotherapies (AABB). There are numerous challenges hindering African countries from supplying quality plasma for fractionation. These include:

- Inadequate or noncompliance with quality management systems and GMP in accordance with WHO or other internationally recognized guidelines: Quality management systems are well-documented; however, the implementation and adherence to their policies is found to be lacking in practice. Similarly, GMP requirements are often understood by staff, but due to inadequate resources, insufficient funding to replenish consumables, poor facilities, lack of schedules for the repair and maintenance of equipment, and deficient procedures, requirements are circumvented, thereby compromising GMP.
- Absence of national policies or regulations allowing for the export of surplus plasma for fractionation into PDMPs: The implementation of a regulation or policy can take several months or years as it requires input from multiple stakeholders and is also dependent on the commitment and prioritization by government. Plasma and PDMPs are not well understood by regulators and policy-makers, and this also delays the drafting of necessary regulations.
- National regulatory authorities do not have the knowledge and experience necessary for inspecting and regulating the national blood establishments and evaluating their compliance with international standards: The national blood establishments rely on independent organizations such as the AfSBT or AABB – which comes at a cost – to conduct accreditation assessments to ensure a minimum standard is achieved at all stages of the whole blood collection process, namely, donor selection, whole blood collection, laboratory testing, blood component separation, plasma freezing, labelling, storage and transport.

¹ This statement derives from the WHO Global status report on blood safety and availability 2021 which can be accessed at <https://iris.who.int/bitstream/handle/10665/356165/9789240051683-eng.pdf?sequence=1>. The report is based on data, up to the year 2018, obtained from the Global Database on Blood Safety (GDBS). Major findings of the report were summarized in a fact sheet published in June 2023, which is available at <https://www.who.int/news-room/fact-sheets/detail/blood-safety-and-availability>. The 2023 fact sheet is based on data obtained through the WHO GDBS from 108 countries for the year 2018. To give a more complete overview of the global situation, data for the year 2017 have been used for 40 countries and data for the year 2015 have been used for 23 countries, where current data are not available. Overall, responses received from 171 countries cover 97.5% of the world's population.

- Limited financial support is available from government to purchase the equipment and consumables required for blood collection and component separation into plasma products: Most blood establishments in LMICs are fully funded by government and do not utilize any established form of cost recovery model. A full or partial cost recovery model allows the blood establishment to sell blood and blood products to both state and private health facilities at prices that cover the cost of blood collection and include a reasonable margin to ensure the blood transfusion service is self-sustaining. Due to the numerous health priorities and limited financial resources in LMICs, national governments are often unable to provide resources to increase whole blood collection, for example, by supplying suitable collection bags to allow for component separation into plasma products and advanced testing platforms to ensure the safety of blood and plasma.
- High staff turnover, inadequate skills and training: Many blood transfusion services in LMICs have reported high staff turnover, which results in staff shortages. Recruitment of adequately qualified and skilled staff is also a challenge, especially in remote or rural areas. Due to high staff turnover and the necessity to recruit new staff, knowledge transfer does not always take place and training of new staff may be delayed, which impacts compliance with GMP requirements.
- Poor transportation infrastructure: Plasma for fractionation needs to be collected, separated and stored at temperatures colder than -20°C within 24 hours to extract labile proteins and 72 hours to extract non-labile proteins. These time requirements are critical to ensure that the plasma will be suitable for fractionation into the appropriate PDMPs. However, in remote and rural areas where road transportation is not well established, delivery of whole blood for separation and plasma freezing within the required timelines poses a significant challenge.

In countries that are landlocked and do not have easy access to seaports, transportation of plasma to international fractionation facilities is a particular challenge as this increases the risk of temperature excursions.

- Inadequate supply of basic amenities such as water and electricity: Availability of clean water and an uninterrupted power supply is also a challenge in LMICs, especially in remote settings. Shortages of water and power will restrict the usage of certain equipment, impact GMP compliance and adversely affect the quality of plasma collected.



Donated plasma as a national resource

Globally, patient needs for treatments with essential PDMPs including clotting factors and immunoglobulin G are not being met. The World Federation for Hemophilia estimates that 70–80% of haemophilia patients, mostly in low-income countries and LMICs, as defined by the World Bank (10), do not have access to adequate treatment (11, 12). Similarly, the International Patient Organization for Primary Immune Deficiencies has reported that, worldwide, a high percentage of patients with primary immunodeficiency do not receive appropriate treatments. These deficiencies in diagnosis and care, which cause otherwise avoidable suffering and mortality, result from failures to ensure the availability and affordability of PDMPs. Lack of appropriate care is further aggravated by failures to diagnose the inherited and acquired disorders that can be treated with these products, often because health care personnel are aware that these treatments will not be available. Validated small-scale preparation of pathogen-reduced cryoprecipitate and immunoglobulin in GMP-compliant blood establishments can provide a first and temporary source of safe plasma protein products for some patients (13). However, sufficient access to PDMPs as the products of choice requires a national commitment to ensuring their availability and affordability, either through humanitarian donations or direct purchasing. Fractionation of domestic plasma under a contract with an internationally accredited foreign manufacturer (contract or toll fractionation) offers an alternative or complementary mechanism to assure a stable supply of PDMPs, in some cases at a net cost lower than that of direct purchasing of products with equivalent characteristics. Through fractionation under a foreign contract, the economic value of plasma as a critical raw material for manufacture of PDMPs is realized. Wastage of plasma that can be fractionated negates its medical and economic value.

A premise of good governance is to maximize the health and fitness of the population. Thus, it follows that providing patient access to essential PDMPs should be a public health goal. The recognition that domestic plasma is a valuable raw material for the manufacture of medically essential PDMPs is central to the goal of achieving a supply of these products sufficient to meet patient needs. This applies to plasma recovered from whole blood, plasma obtained as a by-product of cellular apheresis (“concurrent plasma”) and to plasma prepared directly through plasmapheresis. Discarding “surplus” plasma that is not needed clinically for transfusion or missing the opportunity to generate surplus plasma components from donated whole blood because of the absence of, or a low rate of, separation, therefore contribute to insufficiency of PDMPs in settings where the cost of commercially purchasing them is prohibitive and where this domestic plasma could be fractionated. This is wasteful economically since the value of the surplus plasma

goes unrealized. Additionally, discarding surplus plasma is questionable ethically since it devalues the gifted donation and deprives needy patients of potential access to quality-of-life-enhancing and often life-saving PDMPs.

A government-supported programme that recognizes the medical and economic value of domestic plasma as a resource for manufacture of PDMPs can help to mitigate unmet patient needs by assuring preparation, storage and transport of donated plasma that meets quality standards and a sufficient aggregate volume for foreign fractionation. This can be achieved through establishment of a national blood and plasma programme that:

- sets and enforces standards for plasma preparation;
- coordinates the movement of plasma domestically and across borders;
- negotiates contracts for foreign fractionation; and
- authorizes importation of PDMPs made with domestic plasma or made with foreign plasma but returned in exchange for transferred domestic plasma.

Furthermore, utilization of domestic plasma to secure a reliable supply of PDMPs can avert supply disruptions that can arise from external factors affecting the international availability of plasma for fractionation and the commercial marketing of PDMPs. Viewed in this way, the supply of domestic plasma suitable for fractionation can be recognized as a strategic national resource relevant to public health, which it is critical to protect and maintain, analogous to sources of energy or clean water (14).



4

CHAPTER

Ethical concerns regarding plasma exportation

Various objections to plasma exportation by relevant national decision-makers arise from ethical concerns. These can be addressed through a well-managed programme of cross-border transfer of domestic plasma. Ethics are defined as the “moral principles that govern a person’s behaviour or the conducting of an activity” (15). In the context of this document, the “activity” is the provision of blood and plasma products derived from donated blood and plasma. Classical principles of bioethics (16) applicable to this activity include:

- respect for autonomy: respecting the self-determination and choices of patients and participants;
- nonmaleficence: avoiding or minimizing harm to others;
- beneficence: promoting or doing good for others; and
- justice: ensuring fairness and equality in the distribution of benefits and burdens.

Consistent with these principles, harms to blood and plasma donors must be avoided and benefits to patients must be realized. From the ethical perspective, suboptimal use of donated blood and plasma (failures of respect for donor autonomy and of beneficence), and adverse impacts of plasma exportation on the health system (failures of beneficence and justice) are perceived overtly or intuitively as unacceptable consequences of plasma exportation.

These adverse outcomes, which can and should be avoided, can be understood to arise from exportation of domestic plasma that is not linked to benefits to patients and to the health system in which the donations are made. Ethical acceptance of cross-border transfer of domestic plasma therefore depends on:

- assurance of donor safety;
- optimal use of donated blood and plasma to benefit patients, for example, through linkage to access to essential PDMPs in public contracts, or use of plasma that would otherwise be discarded; and

- management of the blood and plasma system to prevent negative social and economic impacts that might undermine access to safe blood for transfusion.

In this context, donors should be informed that PDMPs manufactured from their altruistic donations will become economically valued products that have costs.

Moreover, it follows that moral duties exist to avoid exploitation of donors and to make optimal use of this donated human substance for the benefit of both:

- the patients who depend on the medicines derived from it; and
- the health system in which the donation is made (17).

These ethically driven objectives can be achieved under nationally or regionally organized plasma programmes that link plasma exportation to importation of PDMPs. Such programmes would have regulatory safeguards in place to protect donor health, to assure quality of the PDMPs and to provide patients with fair access to the product.

Current barriers to cross-border domestic plasma transfer undermine the interest in making optimal use of donated blood and plasma for two main reasons:

- Countries lacking the capacity to fractionate plasma domestically are deprived of the possibility of fractionation agreements that promote enhancements in the domestic blood and plasma collection systems. Upgrading of the blood systems has been taking place in several countries with ambitions to fractionate.
- Countries are unable to utilize and export fractions from the fractionation process that are surplus to their needs. Being able to do so would allow products made from these fractions to be supplied in areas that otherwise lack the capacity to access them.

The development of a suitable framework for plasma exports on a regional level, overseen by a common regulatory framework, would allow the fractionation of plasma by contract. This would enable supplies to countries lacking fractionation capacity or that individually produce an insufficient volume of plasma to permit non-domestic fractionation. This is only possible if blood establishments conform to the relevant international standards. For this reason, access to plasma products obtained through these arrangements will also enhance the quality of the blood service in countries contributing plasma in general.

An adequate supply of plasma for fractionation can only be ensured through the provision of source plasma, augmenting the supply of recovered plasma. The frequency of donation by plasmapheresis presents an ethical challenge because the health of the donor needs to be protected from protein depletion (18). Thus, the frequency and volume of donation should be supported by scientific data to demonstrate adequate protection of donor safety and plasma quality. When these principles are applied, there is no ethical impediment to widening the donor pool through plasmapheresis if adequate standards are in place. On the other hand, collecting blood in excess of red cell needs in order to supply plasma for fractionation, as has been done in the past, raises clear ethical issues related to the wastage of unnecessarily donated red cells and this should be discouraged.

Whole blood collection with preparation of recovered plasma is mainly performed by public sectors, whereas plasma collection by plasmapheresis is typically performed by the private sector, but increasingly also by the public sector in parallel to whole blood collection. Establishing a balance between these coexisting systems is crucial. Otherwise, the supply of blood and blood components for transfusion may be influenced adversely. The national authority with responsibility for blood and plasma collection must be tasked with providing guidance to both sectors when it comes to donor recruitment and donor retention. Assuring financial neutrality in compensation or payment of source plasma donors avoids a potentially negative impact on public willingness to donate whole blood by limiting an incentive that might otherwise favour donation of source plasma over donation of whole blood. Financial neutrality also protects source plasma donors from exploitation and prevents the preferential recruitment of economically disadvantaged individuals.



CHAPTER 5

Legal and policy concerns

Besides the ethical concerns discussed in Chapter 4 and the economic aspects discussed in Chapter 7, there are also legal concerns related to cross-border transfer of domestic plasma and the subsequent import of PDMPs. These apply to the following:

- Plasma needs to be of sufficient quality to be accepted for fractionation.
- Plasma needs to be transported to the fractionator, in a different country.
- PDMPs are to be imported into the plasma-supplying country.

For the first and the second aspects, the specifications of the fractionator need to be adhered to during donation, manufacture, procurement, storage and transport of the plasma. This entails compliance with GMP as laid down, for example, in the *WHO guidelines on good manufacturing practices for blood establishments* (19). These requirements need oversight by the relevant national regulatory agencies.

For the regulatory agencies to be able to fulfil this duty, they need the legal mandate to license and inspect the blood establishments, and, if necessary, to sanction them in cases of misconduct. Another important aspect is traceability between the single donations and the resulting PDMPs to allow follow-up of cases of potential transmission of infectious bloodborne diseases. For this, data about the donations need to be stored safely and to be transferred to the fractionator, for example, in a plasma master file or equivalent documentation. This requirement should be addressed in a national haemovigilance and pharmacovigilance system that allows for the surveillance of the entire transfusion chain (20).

The plasma has to be transported under standardized temperature-controlled conditions, which should be validated and monitored. In the case of cross-border transport, it is necessary to clearly define who is responsible for this control on both sides of the border. Since the transport over the border constitutes an export of goods, corresponding legal requirements need to be in place or to be established to permit this cross-border transport. Moreover, the customs personnel need to be educated and trained accordingly.

In some countries, a legal framework is required to issue permission for plasma export (as well as for plasma import) as it is a human-sourced material. This permission is sometimes issued by the authorities per shipment. From its side, the authority of the country where fractionation takes place must approve the plasma and the centres in which the plasma has been collected, so that the plasma can be allowed to enter the receiving country.

For the third aspect, the legal requirements for the import of the product, including Good Delivery Practices, need to be fulfilled. As for all medicinal products, legal controls are in place to prevent importation of compromised or counterfeit PDMPs. The PDMPs will need to have a current MA in the country to which they are imported. Where the imported PDMPs include non-domestic plasma, a scientific and regulatory assessment has to be made to determine the acceptability of the final products based on comparable risks of the differently sourced plasmas. If a country lacks regulatory standards for authorization of PDMPs, they may be adapted from regulatory systems in more advanced settings. It is also important to have sufficient personnel available who are adequately trained to perform these tasks. Where the MA of PDMPs is still a challenge, a short-term solution might be to base the national authorization on reliance on the MAs of non-domestic internationally recognized agencies, for example, WHO-listed authorities. Since the long-term aim is to supply the country with sufficient PDMPs for the domestic patients who need them, a sole importation in response to the demand of a physician for products to treat a single patient is not a viable national option, although legal requirements may permit this under exceptional circumstances of individual medical need.

In some countries, PDMPs are subject to lot release by a national competent authority (NCA). These will most likely come from the NCA of the country or region where the marketing authority holder (MAH) of the PDMPs is located. If the plasma-supplying country requires lot release certificates, it needs to be clear in which form these are to be issued. For example, the European lot release certificates of the Official Medicines Control Laboratory (OMCL) network are legally valid only within the European Union and the European Economic Area (EU/EEA). However, increasing numbers of countries outside the EU/EEA recognize the value of the Official Control Authority Batch Release (OCABR) certificate and are requesting such a certificate from the manufacturer. Indeed, for these recipient countries, adoption of the EU OCABR certificate is also an added guarantee of the quality of the batches marketed (20, 21). While adoption of the EU OCABR is a possibility, documentation of regulatory requirements for batch release can be met through use of other comparable formats. Regulations and standards need to be adopted mutually between the authorities, and their respective duties should be clearly defined. Among others, the main topics for clarification include duties relating to initial and follow-up GMP inspection, change and deviation control, and complaint and recall procedures.

A detailed quality agreement is required covering all aspects of the contract for plasma and PDMPs. This includes defined roles and responsibilities, use of specified donor screening test kits and facilities, look-back procedures, and provision of epidemiological data. In some cases, this agreement will need to be approved by the local authorities. Establishment of post-marketing surveillance procedures with clear definitions of roles and responsibilities is also important.

These aspects should ideally be addressed in a regulatory framework that is under the control of the NCAs for medicines of both the countries that are providing the plasma (and receiving the PDMPs), and the country where the fractionation and manufacturing takes place. Both agencies need clarity about each other's responsibilities to allow a smooth flow of goods and information.

Therefore, to facilitate the transport of plasma across the border to the fractionator and the reimportation of the PDMPs, an adequate regulatory framework must be developed. There are already good examples of how this can work in regions of the developing world. In sub-Saharan Africa, where the supply of the population with PDMPs is still low despite the waste of plasma, fruitful interactions between the national blood transfusion services and a regional

fractionator are helping to identify the gaps in the regulatory system and to support their remediation. Similar activities are taking place in south-east Asia. Another way forward is to use the WHO Global Benchmarking Tool plus Blood (22) for a self-assessment to test the status of the national regulatory system. Based on the results of the analyses, insufficient regulatory frameworks can be developed and/or strengthened and adapted to the specific situation in each country. This may, for example, be accomplished by making use of the experiences of countries that already export plasma and import PDMPs in return, for example, in the form of collaborations or twinning projects. Also, existing regional organizations may be used or may be formed allowing countries with different competences to complement each other to form a network of reliance. One such example is the ZaZiBoNa-Network for medicines registration (23). Such regional networks may also blend their plasma to reach a sufficient pool size for fractionation.

Countries seeking to collect plasma in a way that facilitates acceptance by fractionators can, as an important first step, adopt internationally recognized standards like those of the Pharmaceutical Inspection Cooperation Scheme (PIC/S) (24) and WHO guidance (4). PIC/S Guidance documents stipulate all quality measures from donation to GMP. *PIC/S good practice guidance for blood establishments and hospital blood banks* (PE 005) was revised in 2021 and is a key document to adopt in local regulations. By adopting PIC/S standards or comparable internationally recognized standards, countries can comply with the requirements of regulatory authorities in countries where fractionation facilities are operated, as well as those specified by the fractionators.



CHAPTER 6

Strategies to protect the health of plasma donors

Ensuring the health and well-being of blood and plasma donors is crucial in all countries and is a particular concern in LMICs. The approach taken should not only put an emphasis on ethical standards but can also help mitigate concerns of the public regarding the potential “exploitation” of donors.

6.1 Considerations for whole blood donors

In environments with limited resources, ensuring the availability of blood and blood products for blood transfusion in patients is often a challenge. Where there is a shortage of blood products, efforts at blood donor recruitment should aim to increase whole blood donations that are also used to prepare recovered plasma.

Where plasma for fractionation is recovered from whole blood, or is a by-product of blood component apheresis, the donor is also the plasma donor, and the donor’s health should be protected by the same means and measures and in line with practices for protecting whole blood donors. Selection and testing of whole blood donors are essential to prevent risks to patients receiving blood transfusions and, equally important, to protect blood donors from adverse outcomes of a donation. Before being selected, donors should fill in a donor questionnaire and undergo a physical examination (pulse, blood pressure and temperature). Laboratory testing, depending on jurisdiction, should include haemoglobin, blood typing and testing for evidence of transfusion-transmitted infectious diseases. Additional donor testing for ferritin level depends on the jurisdiction.

Measures to protect the donor include:

- screening to ensure general health and freedom from certain recent or chronic infections, disorders and exposures;
- determination of a minimum acceptable body weight;
- limitations to the volume and frequency of blood collections;

- performance of the blood collection by well-trained personnel using sterile equipment and aseptic techniques; and
- clinical oversight to monitor for potential hypotensive, allergic or other adverse reactions.

Additionally, donors should be informed of abnormal findings and referred for follow-up medical evaluations and management when appropriate.

An important health focus for whole blood donors is on maintaining the iron store. To address this, ferritin testing can be added to haemoglobin testing as a precautionary measure for the protection of donors. Low iron stores in a donor should lead to an individualized donor interval being set, for example, less frequent donation might be recommended. To protect donors, the donation frequency must be clearly regulated, as is done by the safety rules for substances of human origin (SoHo regulations) in the EU (25, 26) or the regulations of the US Food and Drug Administration (FDA) (27), among others.

In summary, in situations when only recovered plasma is collected, plasma donors' health will be protected mainly in the same way as for any whole blood donor.

6.2 Considerations for plasmapheresis programmes and source plasma donors

In some countries, the blood system has achieved a sufficient level of availability of blood for transfusion that meets appropriate quality standards along with proper oversight of blood collection and component separation. Additionally, they possess knowledge of GMP standards regarding collecting plasma intended for fractionation by recovering it from whole blood donations. For these countries, the next step to be considered is initiating plasmapheresis programmes. These programmes are intended to increase the total volume of domestic plasma suitable for fractionation.

A stringent regulatory environment, as outlined in existing frameworks such as the SoHo or FDA regulations, or in the practices of WHO-listed regulatory bodies, is a prerequisite for policy implementation and oversight to ensure the safety of plasma donors.

Requirements vary by country (28) but often include:

- health screening for potential and current plasma donors according to strict criteria;
- strategies to mitigate the risk of adverse events among specific groups of donors, i.e. deferrals for those with low body weight or low haematocrit;
- limits on how frequently or how much plasma can be donated;
- measurement of total serum protein and immunoglobulin G;
- inspection and certification and/or licensure of facilities collecting plasma;
- mandated staffing of plasma donation centres by licensed medical professionals;

- regular training and professional development for staff of plasma donation centres;
- requirements to track and report any adverse health events associated with a plasma donation and to ensure medical oversight at all times;
- recommendations on limited compensation of plasma donors for reimbursement of expenses incurred and compensation for time and inconvenience, consistent with financial neutrality;
- obligation for donors to be screened and plasma to be tested for infectious diseases on two separate occasions before donated plasma can be manufactured into PDMPs.

Within their well-regulated environment, high-quality plasma collection centres impose a strong safety framework to ensure that plasma donors are and remain healthy. To protect their health, potential donors are required to undergo a series of standard health screenings:

- health history questionnaire (regular medications, recent medical procedures, medical conditions and risks);
- physical examination and vital signs (temperature, blood pressure, pulse, weight, examination of the skin of the arms);
- blood and plasma tests (for example, haematocrit, protein testing, haemoglobin and tests for infectious diseases) (29).

People with a medical condition or other factors that may impact their ability to safely donate plasma are not accepted as donors. However, if the health condition or other factors that would impact a person's ability to safely donate plasma has changed or been resolved, it may be possible for them to donate plasma at a later time. Therefore, the donor may be asked to return after a specific temporary deferral period to be reconsidered as a donor.

The plasmapheresis process (having blood removed, plasma separated, and red blood cells, white blood cells and platelets returned to the donor's body) is monitored throughout by trained and licensed medical professionals (phlebotomists).

- Plasma donors receive oral or intravenous fluids at the end of each donation to address volume loss (30–32).
- In some countries, donors are further monitored for a brief period after their donation.

Additional measures such as those established in the International Quality Plasma Programme (IQPP) of the Plasma Protein Therapeutics Association (PPTA) should be considered. These include independent, third-party evaluation and recognition of a plasma donation centre's adherence to global industry standards for donor health protection and management, which complement regulatory requirements (33). The IQPP includes measures to:

- support plasma donors with information on protecting their health and leading a healthy lifestyle;
- enable plasma to be replenished quickly by providing fluids (34); and
- monitor, manage and record donor adverse events (35) and, when relevant, donor outcomes.

A stringent regulatory environment, a comprehensive safety framework in centres and adherence to voluntary industry standards mean that adverse effects of plasmapheresis are rare. Collection of records of immediate donor adverse events as part of the plasma programme, combined with more data on long-term health, will contribute to safe and sustainable plasma collection (36, 37).

6.3 Haemovigilance

Haemovigilance for donors, or donor vigilance, is a system to register adverse events. Importantly, since adverse events and reactions are infrequently seen, there is a need to record and analyse them. The adverse effects of blood donation are an important indicator for determining the efficacy of blood transfusion safety measures. Such measures include standards for acceptance of the volume of donated blood or plasma in relation to the body mass and environmental or social conditions, for example, ambient temperature, recent fasting, etc. Also, many countries with limited resources are situated in high-temperature zones and their population may have a lower mean body weight than in Europe and North America, which could potentially lead to a higher risk of adverse effects.

Standardized donor-vigilance data on regular donors should be recorded and evaluated to detect long-term undesirable side-effects (38, 39).



Alternative arrangements and economic considerations for fractionation of domestic plasma

7.1 Factors to consider

When establishing a sustainable and efficient system for the fractionation of domestic plasma, it is necessary to explore the alternative arrangements to find the one that best suits the unique needs and constraints of the local health care system and patients' needs.² The decision to adopt a specific arrangement should be informed by a thorough understanding of various critical factors that need to be examined to determine the feasibility and effectiveness of the fractionation programme.

The selection of an arrangement for the fractionation of domestic plasma thus requires a balanced consideration of various factors:

National requirements

- The specific requirements and resources of the local health care system are linked to the unique characteristics and needs of the local health care infrastructure. These include logistics, storage and distribution capabilities. All these factors must be taken into account to ensure that the fractionated plasma products are accessible and beneficial to the intended recipients nationwide.
- The feasibility of having access to and being able to assign trained and skilled personnel representing the key stakeholders to lead and carry out the process of engaging with potential fractionators should be explored. Competencies needed include:
 - knowledge of local plasma collection and the related quality and GMP demands;
 - commercial and contractual knowledge for the contract; and

² Sale of plasma that is not linked to procurement of PDMPs is an economic arrangement that exists primarily in high-income countries. In this arrangement, plasma is sold to a fractionator and the PDMPs are purchased independently on the open market. The purchaser has the option to select products from the full range of available products and gives patients the option to have access to alternative brands of products in the case of intolerance.

- knowledge of the local distribution of, and market for, medicinal products.

If domestic fractionation is an eventual goal, competencies to plan for and operate a domestic facility can be included.

General regulatory aspects concerning plasma and PDMPs

- The maturity of the domestic regulatory system – its robustness and developmental stage – plays a pivotal role in ensuring that the quality of the plasma, the fractionation process and the resulting PDMPs meet international safety and quality standards.
- The regulatory requirements in the country of the fractionator need to allow for use of the domestic plasma from the plasma-providing country.
- The ability to add, or not, domestic plasma to the fractionator's plasma master file (or meet similar regulatory requirements for collecting and ensuring the quality of plasma for fractionation). If this is possible, the PDMPs made from this plasma can be approved for sale by the plasma fractionator.
- Allowance for potential mixing of the domestic plasma with plasma from other sources in product preparation. Mixing may result in marketing of such PDMPs in other countries. This also means that the country supplying the plasma would receive PDMPs made from plasma collected in other areas. Such an arrangement requires careful consideration of compatibility and regulatory compliance.

Product (PDMP)-related

- The PDMPs imported from the fractionator need to have MA that meets the national requirements.
- The pharmacovigilance system for imported products needs to be in place according to local regulations.
- The suitability of the dosage form of PDMPs to be administered is crucial for their efficacy and acceptance within the local health care context (for example, capacity to manufacture formulations for paediatric use). The selected fractionation arrangement should, therefore, align as well as possible with the dosage and pharmaceutical form preferences and needs of the local population in the importing country.
- The plasma product range, in terms of the diversity of PDMPs that can be produced and distributed, is essential for comprehensively addressing the health needs of the community. The fractionation arrangement should facilitate a product range that matches the local demand for PDMPs, although this can also be achieved in a stepwise fashion. It is important to acknowledge that sometimes additional imported PDMPs will be necessary to meet the needs of certain patients for specific PDMPs.

Technical and quality considerations

- The volume of plasma available directly impacts the choice of fractionation arrangement, as it influences the scale and feasibility of operations of many global fractionators, and the range of PDMPs and dosage forms that can be made from the plasma.

- The contract with a fractionator will include some form of quality agreement where the details of the plasma for fractionation and many other quality-related aspects can be described in detail.

Economic considerations

- Toll or contract manufacturing: the cost of fractionation as well as the economic implications of different fractionation methodologies should be evaluated to ensure that the chosen approach is financially viable and sustainable in the long term. It is important to inquire into the overall yield of the different PDMPs as this can have an economic impact, favouring processes with a high yield. More details on consideration of the economic impacts of a toll or contract manufacturing arrangement are discussed as an example in section 7.3.
- Where arrangements including sales of plasma are linked to importation of PDMPs, the prices of these need to be agreed.
- Fluctuating currency exchange rates and tariffs (as seen during the COVID-19 crisis when there was a substantial increase in the cost of transportation) have to be considered in any arrangement. Fluctuating rates can affect the overall cost of the export–import cycle, and restrictions on tariffs or trade can add unforeseen costs to the process.

7.2 Possible alternative arrangements

A multifaceted approach, encompassing both the strategic factors influencing the choice of fractionation arrangements and the potential models that could be implemented, can be adopted. Potential alternative arrangements to ensure a strategic supply of PDMPs encompass:

- *Contract or toll fractionation.* This refers to the situation where domestic plasma meeting the criteria for fractionation – as verified by national regulatory authorities – is collected and shipped for processing by a fractionator, typically located abroad. The resulting PDMPs are subsequently returned to the country of origin under contractual terms. This model is suitable for countries lacking domestic fractionation capacity. Under such a contract, and assuming compliance with regulatory requirements, part of the intermediate fractions or of the PDMPs produced, which are not needed domestically to meet the local needs of the country supplying the plasma, may be kept by the plasma fractionator and distributed in other countries or regions. A contract or toll manufacturing arrangement can also be designed as an intermediary step prior to a transfer of technology from the fractionator to a suitable partner in the country seeking to invest in domestic fractionation of plasma and supply of PDMPs.
- *Sales of plasma to a fractionator clearly linked to purchase of PDMPs from the same fractionator, within the same contract.* In this scenario, the purchased PDMPs can be manufactured from the same plasma as is sold to the manufacturer, but there are also arrangements where the plasma is sold to the fractionator and the PDMPs are also made from plasma (in volumes that have been agreed upon) from other approved plasma sources. This type of contract has many of the same elements as the above-mentioned contract/toll manufacturing one, but a different economic structure.
- *Public–private partnerships.* These partnerships are collaborations between governments and private entities. They are designed to organize a contract plasma fractionation programme or to establish local fractionation

facilities in a system attempting to combine public oversight and private efficiency. Under such arrangements, a joint venture between the fractionator and the national ministry of health may open a number of plasma collection centres. These may be operated by the fractionator and collect plasma to be used for fractionation outside the country. Later, when possible, fractionation may be undertaken domestically, for provision of PDMPs to the population.

- *Regional fractionation hubs:* This is a cooperative model where several countries pool resources and plasma, demonstrated to be of equivalent quality and safety, to establish a shared fractionation facility. This would optimize resources, training requirements, and economies of scale and facilitate access to PDMPs across borders. Such a model would require investment in a fractionation facility for this purpose.
- *In-country fractionation facilities.* This is where a domestic infrastructure ensures local fractionation capabilities, enabling domestic control over the production and supply of at least parts of the PDMPs. Establishing a domestic facility, when justified, is usually best achieved after a ramp-up phase of contract fractionation that allows fast-track access to needed PDMPs and can serve as a critical phase of increasing plasma production and training of local operators.
- *Hybrid models:* a tailored approach may combine elements of the above-mentioned strategies to meet specific domestic or regional needs and capabilities.

Any arrangement for plasma transfer aimed at production of PDMPs needs dedicated and knowledgeable teams and suitable stakeholders for both partners. Public–private partnerships, where collaborations between governments and private entities are designed to organize a contract plasma fractionation programme or to establish local fractionation facilities in a system attempting to combine public oversight and private efficiency, may be one option.

The optimal arrangement for the fractionation of domestic plasma is thus contingent upon a comprehensive evaluation of both the strategic factors at play and the practical models available. By carefully considering the regulatory, economic and health care landscape, alongside the potential fractionation strategies, local stakeholders, in partnership with foreign entities and organizations, can devise a plan that maximizes the use of domestic plasma resources. This plan should enhance the availability, accessibility and quality of PDMPs, under appropriate control of the national health care system, thereby addressing the health needs of the local population effectively and sustainably.

7.3 Example of economic considerations in a contract or toll fractionation agreement

In many cases of contract or toll plasma fractionation, the plasma is not actually sold to the fractionator. Instead, the arrangement involves a service contract where the fractionator processes the plasma into PDMPs and the manufactured PDMPs are returned to the plasma supplier. In this setup, invoicing does not cover the cost of the plasma itself but instead is based on the production services required to convert plasma into PDMPs. The economic terms for this service, which should be clearly defined in a legal and technical contract, can be based on a fee per litre of plasma fractionated with an agreed and guaranteed yield of the end-products. Alternatively, it may be based on a fee per unit (IU or g) of each PDMP actually returned to the plasma supplier. The lack of invoicing for the plasma itself focuses the practical economic exchanges at the time of plasma exportation towards service costs, logistics and quality assurance, also

playing a role in the financial viability of a contract plasma fractionation programme. The invoicing for the fractionation service is the main critical component in the overall economics of the arrangement. It must be accurately calculated to include not just the direct costs of fractionation but also the indirect costs related to the plasma fractionator's profit, and costs of maintaining the plasma's quality throughout the entire cold chain and transportation process.

Since, in a traditional plasma fractionation agreement, the plasma is not being sold, the primary financial exchanges when plasma is exported encompass the cost of shipping the plasma to the fractionator and of shipping the processed products back to the country of origin. One primary cost element in plasma exportation for contract or toll fractionation is associated with maintaining the cold chain. It is necessary to ensure and record that plasma remains at below-zero temperatures during its storage in a centralized warehouse (typically -20°C or colder) in the country of plasma collection and throughout transit (typically -18 to -20°C or colder), regardless of the distance to the fractionation facility. Specialized refrigerated containers and vehicles are required, which adds to the initial capital and operational costs. Depending upon the locations of the plasma suppliers and the fractionator, it may be feasible to use trucks for shorter distances and relatively small plasma volumes. Shipping containers via boat are usually selected for overseas transportation and for larger volumes but need careful timing and port clearances. Air transport may be selected for small volumes or in specific circumstances and is the most expensive option. It should be kept in mind that any delays or inefficiencies in the selected transport system may compromise the plasma quality and incur financial losses.

Finally, once the plasma has been fractionated, the end-products must be shipped back under conditions in accordance with suitable Good Distribution Practices for medicinal products. This includes temperatures compatible with the demonstrated stability of the PDMPs to guarantee their safety and efficacy. This adds another layer of cost.

In summary, the economic concerns pertaining to the exportation of plasma for contract fractionation include the contractual fee for processing plasma into PDMPs, the cost for cold chain maintenance during storage, transportation logistics and compliance with trade regulations.



Conclusion and recommendations on cross-border transfer of domestic plasma to obtain PDMPs

In LMICs, absent or insufficient access to essential PDMPs, including clotting factor concentrates, polyvalent immunoglobulin G and albumin, compromises the health and survival of patients with inherited clotting disorders, primary immunodeficiency and acute massive haemorrhage. Likewise, although not extensively addressed in this document, limited access to hyperimmune immunoglobulins derived from plasma with a high concentration of specific antibodies compromises patient care. In particular, insufficient access to anti-D immunoglobulin and hepatitis B immunoglobulin compromises prevention of haemolytic disease of the newborn and of hepatitis B virus transmission, respectively. At the same time, there is considerable wastage of domestic plasma that could be used to manufacture PDMPs if it met the quality standards for fractionation. In this context, an organized national or regional programme for fractionation of domestic plasma under a contract with an established foreign manufacturer for reciprocal importation of PDMPs made wholly or in part from that plasma (contract or toll fractionation) is recognized as an effective strategy to improve the availability of essential PDMPs to meet identified patient needs.

Implementation of contract or toll fractionation requires a comprehensive domestic blood programme including regulatory oversight to assure an adequate and stable supply of plasma. The plasma also needs to meet internationally recognized standards of GMP for blood and plasma collection, testing, processing, storage and transport. Ultimately, contract or toll fractionation necessitates cross-border transfer of domestic plasma that is linked contractually to regulatorily authorized importation of PDMPs provided by the manufacturer. In addition to the inherent quality, safety and logistic issues related to management of the transferred plasma, ethical, scientific, legal, regulatory and economic concerns can and should be addressed within the programme to assure its societal acceptance and economic sustainability.

In consideration of the need to improve access to PDMPs in many LMICs and some HICs, WHO recommends removal of national or regional legal and/or policy barriers to cross-border transfer of domestic plasma that is designated for contract or toll fractionation under the following conditions:

- A competent national governmental authority provides oversight to assure that GMPs are followed at all levels of the blood and plasma system.

- Regulatory safeguards are in place to assure the health and safety of donors, including limitations on the volume and frequency of collection of blood and plasma.
- Legal requirements backed by adequate enforcement mechanisms are in place to prevent importation of compromised or counterfeit PDMPs.
- If PDMPs imported under the programme involve use of non-domestic plasma solely or in a mixture with domestic plasma, a scientific and regulatory assessment has been made of the acceptability of the final products, based on comparable risks of the differently sourced plasmas.
- Where a programme of plasmapheresis exists to generate source plasma, measures are in place to avoid compensation in excess of financial neutrality.
- The value of the plasma that is exported as a raw material partly offsets the costs of the finished PDMPs.
- Donors are informed that PDMPs manufactured from their donations that have costs of preparation and procurement will become economically valued products for some patients' health and survival.
- A national programme exists to assure equitable patient access to the PDMPs obtained through contract or toll fractionation.
- Cost reimbursement for procurement of PDMPs under contract or toll fractionation is managed under a sustainable national or regional budgetary commitment.



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